

Solar Power Plant Investment in Indonesia: A decarbonized electrical supply vision to 2040

(1) Introduction

Indonesia has an ambitious target of reaching net zero by 2060 and becoming an advanced economy by 2045 (IEA, 2022a). It has a long way to go as its GDP per capita is 30% lower than the world average and its energy sector remains highly dependent on fossil fuels (IEA, 2022a). As of 2022, coal accounted for 71% of the country's energy consumption and 66% of its electricity consumption (IEA, 2022b).

As illustrated by **Figure 1**, renewable electricity generation in Indonesia is dominated by hydrothermal and geothermal generation. Solar PV and wind currently constitute tiny fractions of renewable electricity generation. **Figure 2** shows the location of existing, planned and under construction renewable power plants in Indonesia. Although few large solar power plants are currently generating in Indonesia, the green squares in the map show that many are planned for the future.

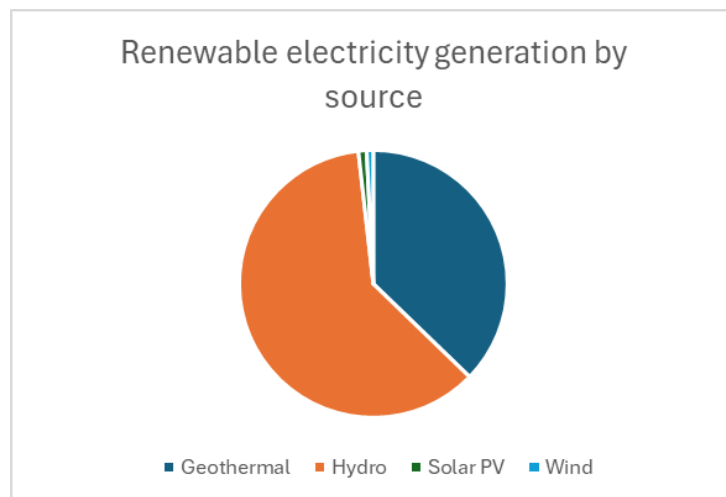


Figure 1: Indonesia renewable electricity mix in 2022 (IEA, 2022b)

Planned, existing and under construction renewable power plants in Indonesia

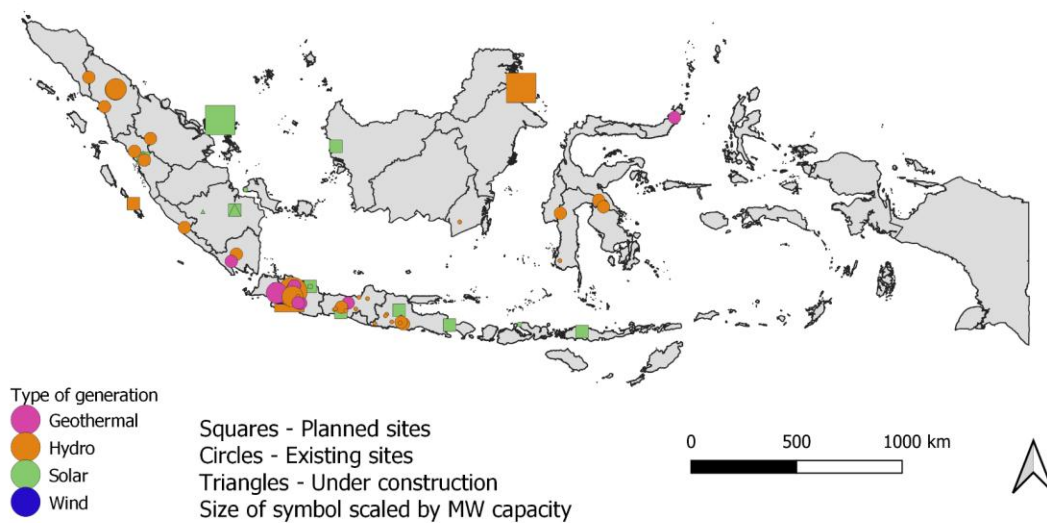


Figure 2: Location of existing, planned and under construction renewable power plants in Indonesia (Zhang, 2024)

The aim of this project is to find areas suitable for additional solar power plants in Indonesia by 2040. To do this, we need an estimate for how much electricity demand there will be in 2040. As demonstrated by **Figure 3**, GDP is a strong predictor of electricity demand.

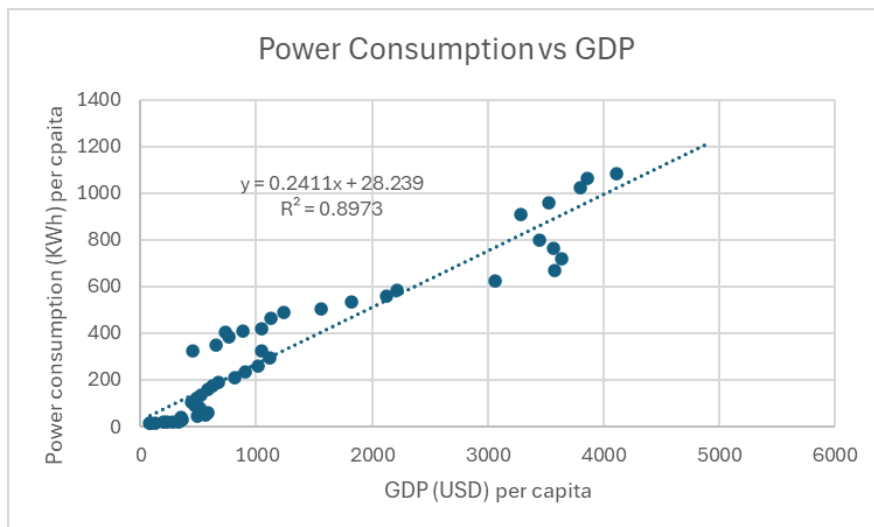


Figure 3: Scatterplot and linear line fit for GDP per capita and power consumption per capita between 1971-2019 (World Bank, 2024)

By 2040, Indonesia's population is projected to reach 312 million people (World Health Organization, 2024). Meanwhile, Indonesia's GDP is forecast to grow at 5% until 2027 and then at 4% until 2040 (Australia Government Department of Foreign Affairs and Trade, 2024). This would result in a GDP per capita in 2040 of \$8895. Plugging this into the above linear fit equation, $y=0.2411x+28.239$, results in a 2040 projection for power consumption per capita of 2.173MWh. With a population of 312 million, the final projection for total electricity consumption in 2040 is 678TWh.

If all power plants were running all the time, this would require generating capacity of 678TWh / (365 * 24) = 77.4GW. We will assume renewable energy plants will supply 38% of demand with a capacity factor of 0.24 and conventional power plants will cover the remaining 62% of demand with a capacity factor of 0.45. Therefore:

$$(0.38c \cdot 0.24) + (0.62c \cdot 0.45) = 77.4\text{GW}$$

$$0.3702c = 77.4\text{GW}$$

$c = 209\text{GW}$, where c represents total capacity required given above capacity factors

Renewable energy plants will supply $0.38 \cdot c = 79.4\text{GW}$. The capacity of currently existing, planned and under construction renewable power plants add up to 19.0GW. This means that the shortfall in 2040 is $79.4\text{GW} - 19.0\text{GW} = 60.4\text{GW}$. We will assume that this shortfall in its entirety will be covered by new solar power plants and the next step is to find suitable locations for these plants.

(2) Literature review

A number of factors need to be considered in order to find suitable sites for new solar power plants, including weather, land use and existing infrastructure. Taking all these factors into consideration requires multi-criteria decision making (MCDM). One of the most widely used MCDM techniques within geographic information systems (GIS) research is the analytical hierarchy process (AHP), proposed by Thomas L. Saaty (Saaty, 1987). In AHP, the factors or criteria are compared pairwise and given weights according to their relative importance. These weights can then be used to produce a suitability index which ranks areas from least suitable to most suitable.

Al Garni and Awasti (2017) used AHP to find that suitable lands for solar power plants in Saudi Arabia followed a pattern of proximity to main roads, transmission lines, and urban cities. Lamichhane (2022) used AHP with a combination of quantitative and qualitative criteria to produce suitability maps for settlements in Bharatpur Metropolitan City, Nepal. Ebru et al (2020) included factors such as solar energy potential, roads, energy transmission lines, transformer centers, slope, aspect, location of dams and rivers, natural gas pipelines, fault lines, land cover and residential areas in their AHP study of site selection for solar power plants in Malatya, Turkey.

A fuzzy analytical hierarchy process (FAHP) is an extension of AHP, also widely used in the literature, which integrates fuzzy set theory to handle uncertainty and imprecision. While AHP

relies on concrete numerical comparisons between criteria, FAHP can handle “fuzzy” preferences such as “criterion A is moderately more important than criterion B”.

Sanchez-Lozano et al (2016) used FAHP to obtain the weights for evaluating available sites for onshore wind farms in Southeastern Spain, and Noorallahi et al (2016) used FAHP to obtain a ranking of the 50 best districts for exploitation of solar energy in Iran.

This project will utilize AHP to find suitable sites for solar power plants in Indonesia. A comprehensive AHP study for solar site selection in Indonesia has already been performed, by Ruiz et al (2020), however that analysis only looked at the West Kalimantan Province of Indonesia. This study will not use as many AHP criteria as Ruiz et al (2020) but will consider all of Indonesia to get a big picture view of where solar plants would be particularly suitable.

(3) Methods and Data

The workflow for this study is the following: obtain data for landscapes, protected areas, population, existing infrastructure and solar radiance -> identify areas which can be excluded as potential power plant sites (constraints) -> use AHP to rank remaining areas according to suitability factors and identify areas best suited to build 60.4GW of new solar power plants.

Constraints: One of Indonesia’s main priorities is to minimize its rapid pace of deforestation (UNFCC, 2022). Therefore, any areas with tree cover were classed as unsuitable for new power plants. Protected natural areas, water bodies, urban areas and populated areas were also classed as unsuitable new large-scale solar power plants.

Suitability factors: The obvious main suitability factor is solar radiance. In addition, proximity to the electrical grid was considered as a suitability factor, since a lower distance to the grid means a lower network connection cost. Finally, proximity to populated areas was considered as a suitability factor. Although this study defines populated areas as unsuitable for new power plants, being close to a populated area is a good thing due to lower electricity transmission cost. Solar power plants located far away from consumers will lose more electricity during transmission than those located closer to people.

Figure 4 and **Figure 5** below show a breakdown of the constraints and suitability factors used for this study and the dataset obtained for each.

Constraints	Data description	Data source
Land cover	Land cover map at 300m resolution with codes for different land cover types. Water bodies (210), urban areas (190), tree cover (50, 62, 80, 100, 110, 160, 170) defined as unsuitable.	ESA (2020)
Populated areas	Binary raster with values 0 or 1 indicating populated areas. Populated areas defined as unsuitable.	WorldPop (2020)
Protected areas	Map of protected areas in Indonesia. All protected areas defined as unsuitable.	UNEP-WCMC and IUCN (2024)
Mountains	30m resolution altitude data. Areas higher than 1500m defined as mountains and therefore unsuitable. Any area within 500m of a mountain point also defined as unsuitable.	Earth Resources Observation and Science (EROS) Center (2018)

Figure 4: Datasets used to identify unsuitable areas for solar power plant site selection

Suitability factors	Data description	Data source
Solar radiation	Average surface solar radiation downwards for hours 08:00, 12:00, 16:00, 18:00 across 10 January, 10 April, 10 July, 10 October	Muñoz Sabater, J. (2019)
Proximity to electrical grid	Map of Indonesia's power lines.	OSM (2018)
Proximity to population centers	Same data as for populated areas in constraints.	WorldPop (2020)

Figure 5: Datasets used to obtain suitability factors for solar power plant site selection

(4) Solar Plant Suitability Analysis

The first step of the solar plant suitability analysis is to exclude unsuitable areas: forests, water bodies, protected areas, urban areas, populated areas and mountain areas. **Figure 6**, **Figure 7**, **Figure 8** and **Figure 9** visualize the excluded areas.

Indonesia protected natural areas

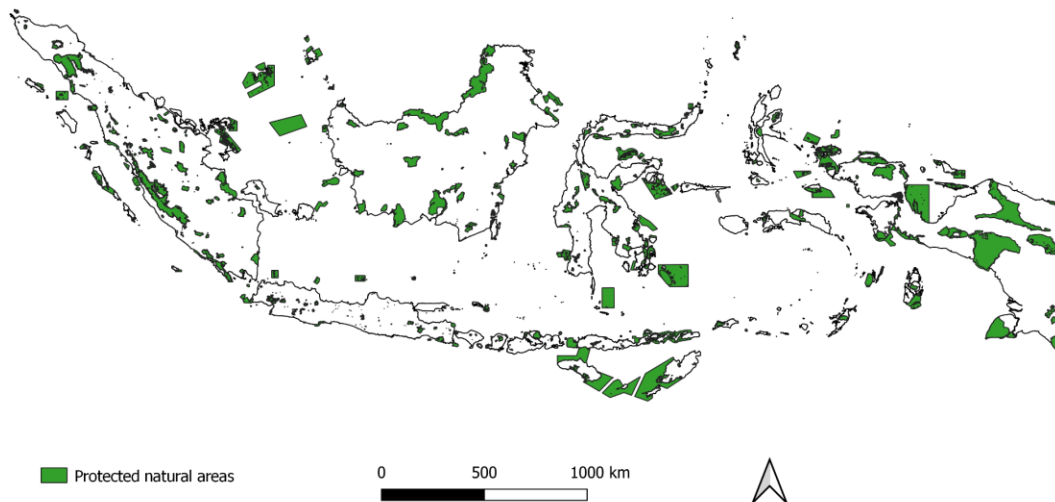


Figure 6: Protected natural areas excluded from analysis (Protected Planet 2024)

Indonesia land use classification

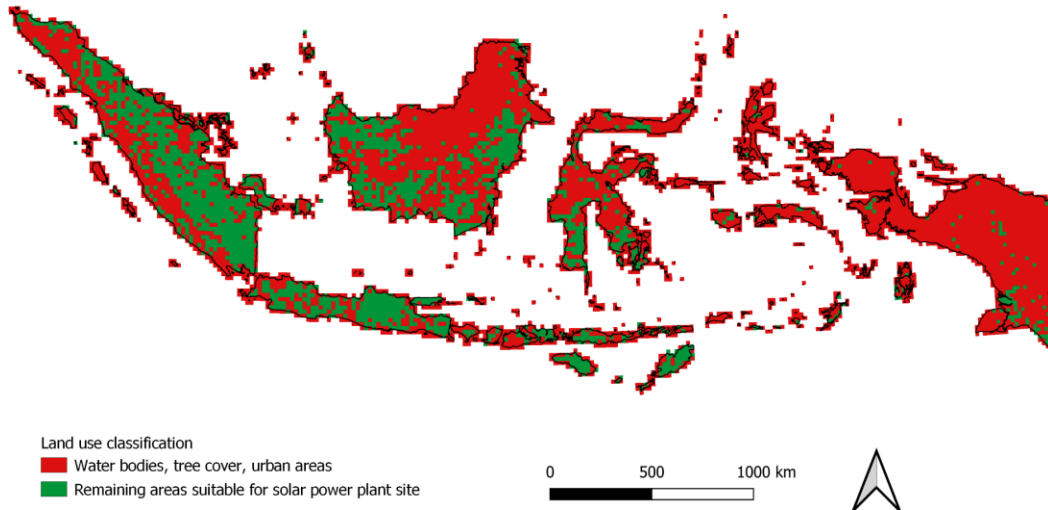


Figure 7: Forests, water bodies and urban areas excluded from analysis (ESA 2020)

Indonesia populated areas

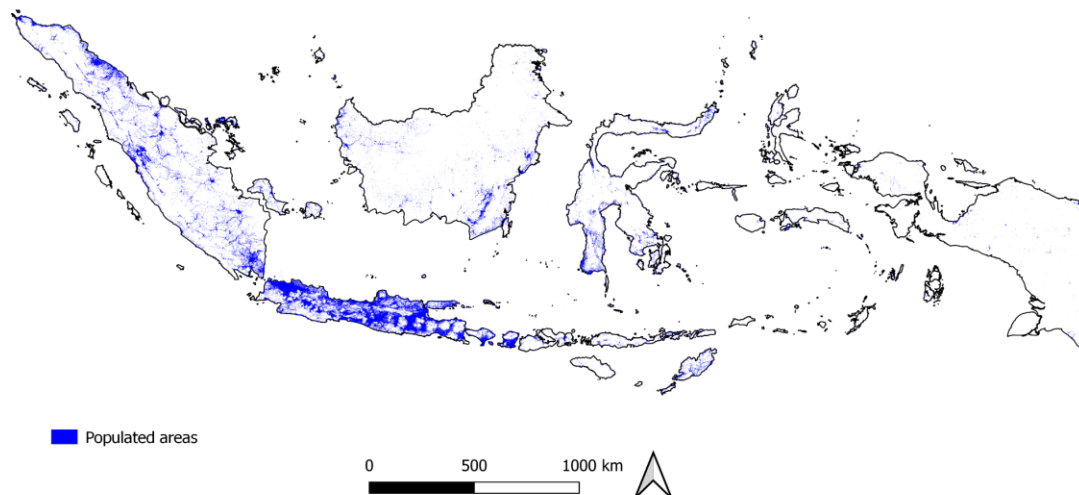


Figure 8: Populated areas excluded from analysis (WorldPop 2020)

Indonesia mountain areas

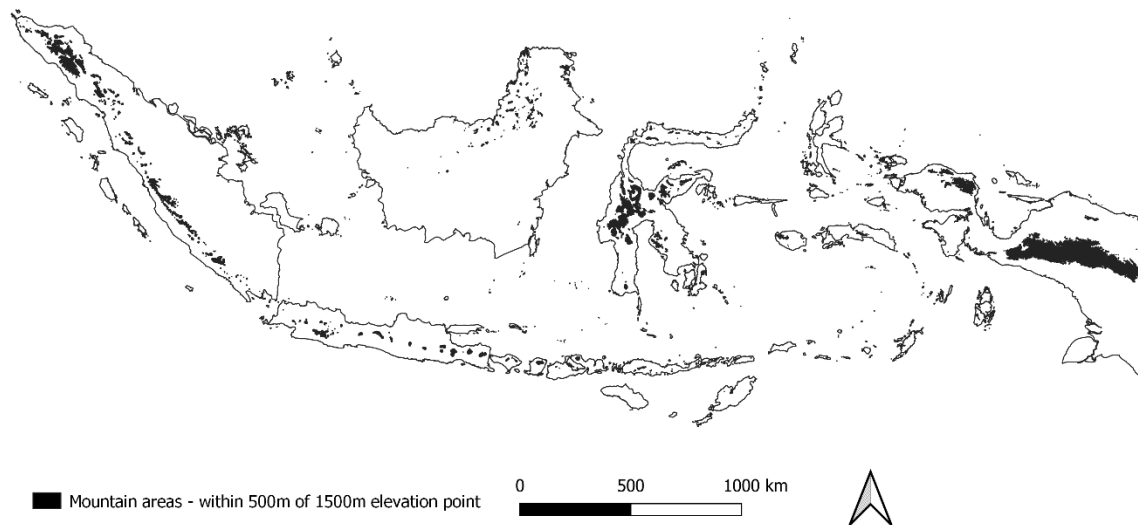


Figure 9: Mountain areas excluded from analysis (EROS 2018)

Once unsuitable areas have been excluded, the AHP process can begin by analyzing our suitability factors. This study uses three suitability factors: solar radiation, proximity to the grid and proximity to populated areas.

For solar radiation data, this study uses ERA5 data for surface solar radiation downwards (SSRD) which represents incident solar shortwave radiation. To get representative day-time solar radiation data across Indonesia, SSRD values were averaged across 4 hours and 4 days: 8am, 12pm, 4pm and 8pm on 10 January, 10 April, 10 July and 10 October.

SSRD values were then converted to units of kWh/hour/m² by the following formula:

$$E \text{ (kWh)} = \text{SSRD} * \text{Area} * \text{Panel efficiency (0.175)} * \text{Performance ratio of system (0.6)}$$

An issue with the ERA5 data is that there is missing data. Spatial interpolation was therefore used to fill in these missing values. The spatial interpolation method used was inverse distance weighted (IDW) interpolation, which estimates values based on neighbors such that the influence of a neighbor is negatively correlated with distance. This is based on the Tobler's first law of geography which states that "Everything is related to everything else. But near things are more related to distant thing" (Tobler, 1970). The k-value in IDW determines how strongly distance influences the interpolation weights and was set equal to 2 in this study.

The result is an interpolated raster map with resolution 20,000 such that each raster cell represents an area of 20km x 20km (400km²) and contains a solar radiation value in units of kWh/hour/m².

Indonesia surface solar radiance

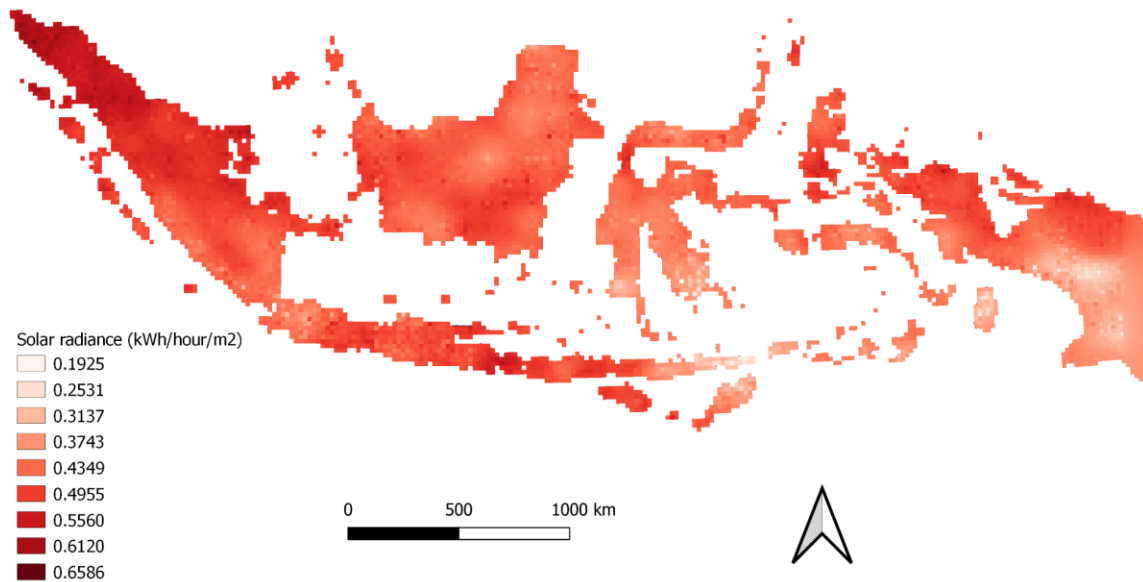


Figure 9: Solar radiance across Indonesia (Muñoz Sabater, J. 2019), missing values filled in by IDW interpolation

The second suitability factor is proximity to the grid. Indonesia's electrical grid is 29,955km long, excluding disused grids. For each raster cell, the distance from the middle of the cell to the middle of the nearest cell containing a grid connection was calculated. For cells containing a grid connection, a default distance of 5000m, 5km, was set. **Figure 10** below shows the location of Indonesia's electricity grid.

Indonesia's electricity grid

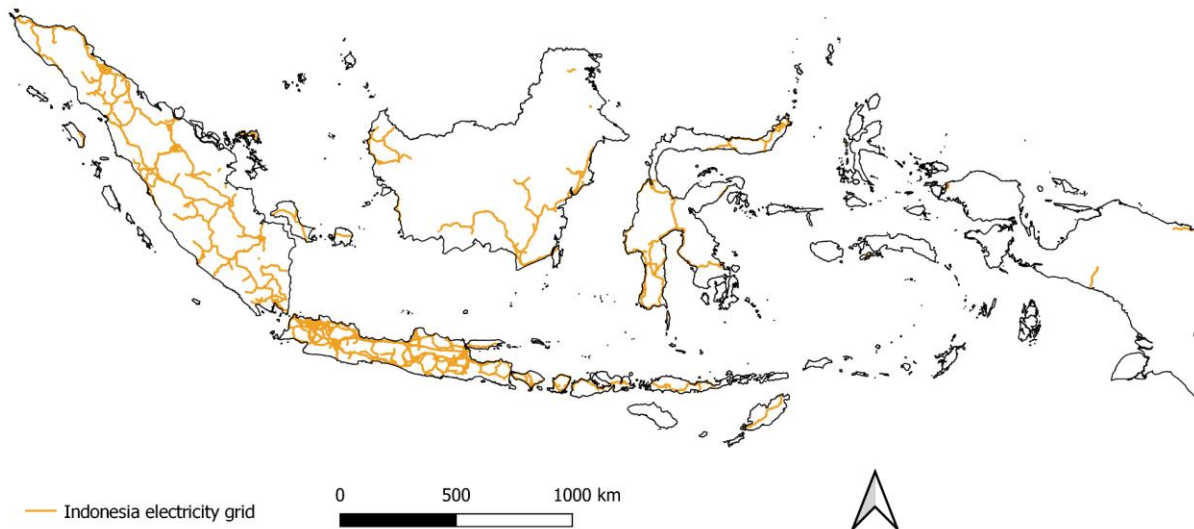


Figure 10: Location of Indonesia's electricity grid

The third suitability factor is proximity to populated areas. The binary population raster was resampled to fit a 20km x 20km resolution. Each cell in the resampled raster contains a value between 0 and 1 which indicates what percentage of the cell is populated. Cells with a population value higher than 0.5 were defined as populated and the distance from each cell to the nearest populated cell was calculated.

Indonesia resampled population

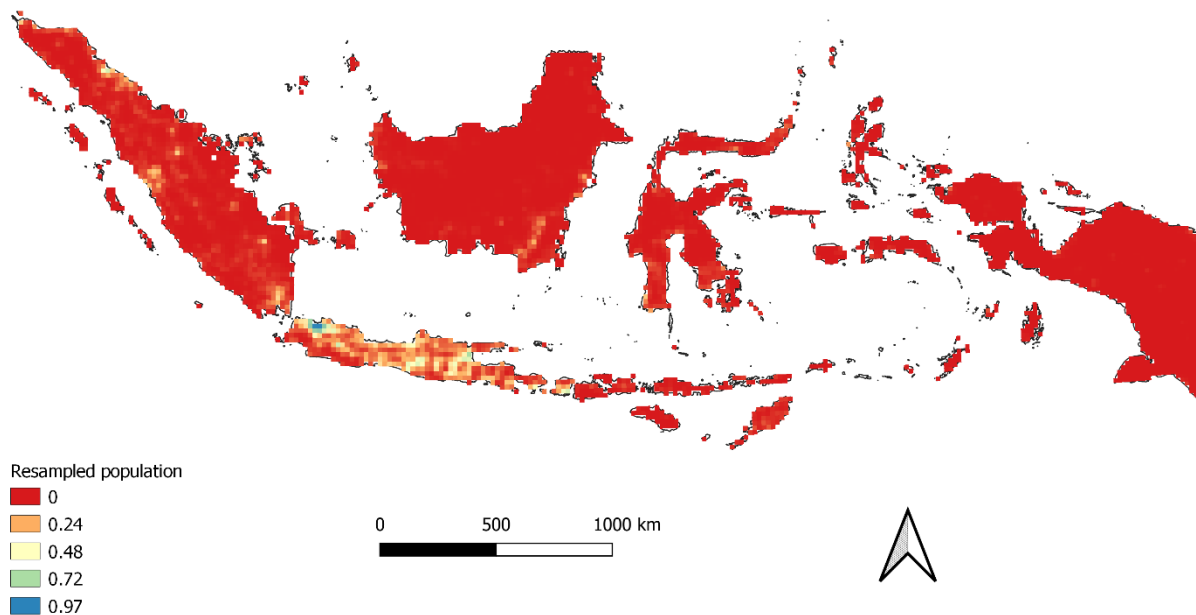


Figure 11: Resampled population (WorldPop 2020)

Among the three suitability factors, solar radiation is defined in this study as three times more important than grid proximity and 5 times more important than population proximity. Grid proximity is defined as twice as important as population proximity. **Figure 12** breaks down our initial AHP weights.

	Solar radiation	Grid proximity	Population proximity
Solar radiation	1	3/1	5/1
Grid proximity	1/3	1	2/1
Population proximity	1/5	1/2	1
Sum	1.533	4.5	8

Figure 12: Pairwise comparison matrix of suitability factors

The final weights are obtained by normalizing the comparison matrix and taking the average of each row:

Normalised	Solar radiation	Grid proximity	Population proximity	Average
Solar radiation	0.65	0.67	0.63	0.65
Grid proximity	0.22	0.22	0.25	0.23
Population proximity	0.13	0.11	0.13	0.12

Figure 13: Normalized pairwise comparison matrix, average column represents the final AHP weights

The next step is to scale each criterion. This means dividing the raster cell areas which have not been excluded from the analysis into five equal quintiles for each criterion. The result is that each raster cell gets a score from 1 to 5 for solar radiation, grid proximity and population proximity.

The final AHP weights are then applied so that each raster cell gets a final suitability score. If a cell has a solar radiation score = 5, grid proximity score = 4 and population proximity score = 3, its final suitability score is: $(5 \times 0.65) + (4 \times 0.23) + (3 \times 0.12) = 4.53$

Each raster cell is then ranked according to suitability score. This study will assume each solar power plant is 12km². We will also assume that only 3% of each 400km² raster cell can be used to develop a solar plant due to real-world complications such as land-rights and unforeseen circumstances. This means that each raster cell will represent one potential new power plant.

Figure 14 shows the suitability score for each raster cell which has not been excluded as unsuitable.

Indonesia suitability score

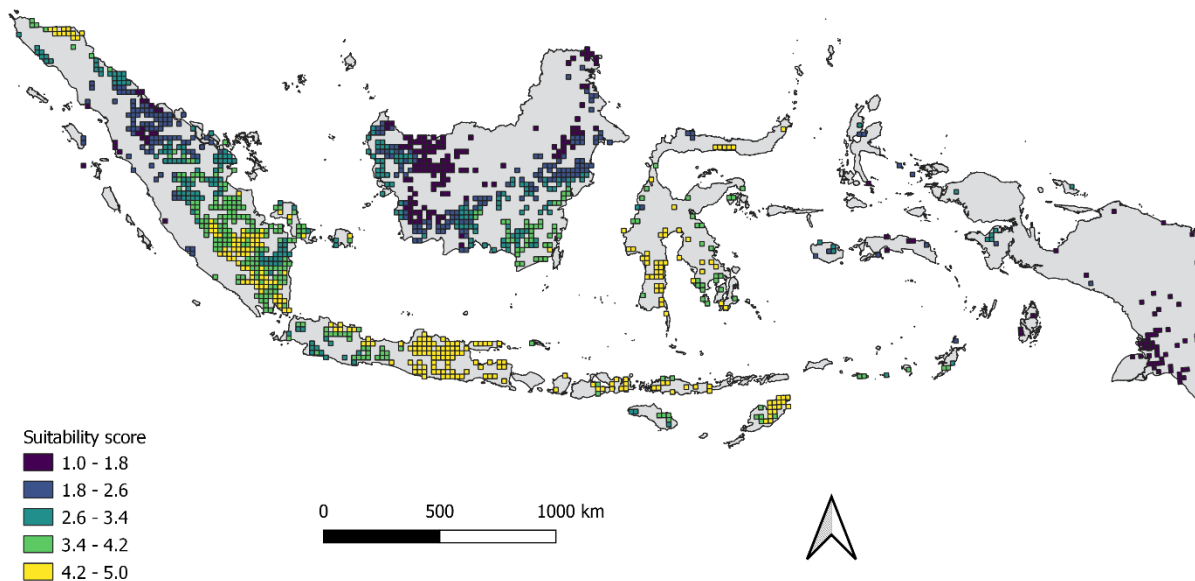


Figure 14: Suitability scores for all raster cells which have not been excluded

Yellow cells in the above map indicate highly suitable areas for solar power plants and we can see that large parts of Indonesia come under this classification in this analysis. In fact, 65 out of 1226 cells have a perfect suitability score of 5.

In the introduction we found that new power plants need to deliver 60.4GW in 2040, which with a capacity factor of 0.24 equals 127TWh. To identify the sites required, we therefore pick the top ranked raster cells which together make up 127TWh of capacity. It turns out that only nine cells are required for this. We therefore sort the 65 cells with perfect scores by solar radiation values and find the following nine optimal locations for nine 12km² solar power plants:

Most suitable power plant locations

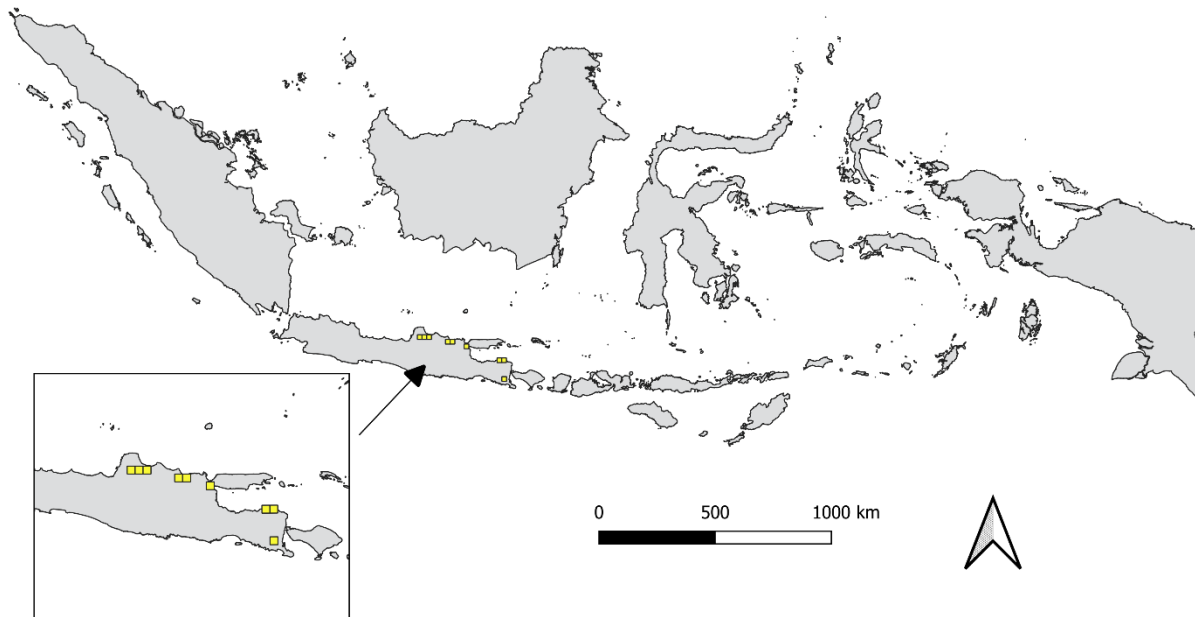


Figure 15: Locations identified as most suitable for solar power plants in Indonesia

(5) Financial Analysis

We will assume that the government can sell electricity to electricity suppliers in 2040 at a price of £0.0767/kWh. The government's yearly revenue from these solar power plants is therefore $127\text{TWh} \times £0.0767/\text{kWh} = £9.7\text{bn}$

The capital expenditure (CAPEX) of the project includes the funds required to acquire, upgrade and maintain physical assets. The main components of this cost are the cost of installing the solar power plants and the cost of connecting them to the network. The project's CAPEX can therefore be determined by the following formula:

$$\text{CAPEX} = (\text{installed capacity} * \text{capacity installation cost}) + (\text{distance between plant and the nearest grid} * \text{network connection cost})$$

We will assume a capacity installation cost of \$1.16mn/MW installed power, a network connection cost of \$590/MW*km, and \$1=£0.8, resulting in a CAPEX of £56.2bn.

Net present value (NPV) measures the current value of future cash flows, discounted to the present using a discount rate. If we assume the solar power plants have a lifetime of 25 years, no operational expenditure (OPEX) and a discount rate of 5%, then the NPV of the project is £81bn – this is higher than 0 and therefore supports the investment opportunity. All other assumptions staying equal, the NPV only turns negative at a discount rate of 17%, which is very high and therefore supports the viability of the investment opportunity.

Finally, the levelized cost of energy (LCOE) is a measure of the average cost per unit of electricity generated over the lifetime of a power plant – it takes the lifetime costs and divides by total energy produced. Using the same assumptions as above, the LCOE comes out as £0.03/kWh – less than half of the assumed selling price of £0.0767/kWh. Increasing the discount rate to 8% gives a slightly higher LCOE of £0.04/kWh, which is still well below the assumed selling price. As with the NPV, the LCOE would only equal the assumed selling price at a discount rate of 17%.

(6) Conclusions

The result of this analysis has been a demand projection for new solar power plants in Indonesia by 2040 and the identification of suitable areas for these new power plants. We have found that nine 12km² power plants can meet the projected demand and we have identified suitable locations for these plants. These locations have high solar radiance values, are close to the electrical grid to minimize network connection costs and are close to populated areas to minimize transmission costs.

The method for selecting suitable sites was based on excluding unsuitable sites and then ranking areas based on a AHP process with solar radiance, proximity to populated areas and proximity to the grid as suitability factors. IDW interpolation was used for spatial filling in missing solar radiance values. A key finding is that large parts of Indonesia are highly suitable for solar plants, beyond the nine locations identified, indicating vast potential for solar power in the country.

Furthermore, we have found that investment in these power plants is likely to be profitable, with a positive NPV and with a low LCOE, under the assumptions detailed in the Financial Analysis section. Discount rates would need to be as high as 17% for the investments to be unprofitable under our OPEX and CAPEX assumptions.

Further research on this topic could use more suitability factors to further distinguish between the highly suitable areas. A more granular raster resolution could also be implemented so that suitable areas can be located with higher precision. Finally, OPEX and CAPEX assumptions could be investigated further for added confidence in the financial analysis.

Below is an appendix of the assumptions made in this study and a link to the code used.

(7) Appendix

Technical specifications assumptions			
Source	Concept	Base assumption	Used assumption
Solar (PV)	Area	Estimated by student based in model	12km ² per solar power plant
Solar (PV)	Yield	0.175	0.175
Solar (PV)	Pr	0.6	0.6
Solar (PV)	Installed capacity	Estimated by student based in model	60.4GW
Renewable energy financial analysis			
Source	Concept	Base assumption	Used assumption
Solar – NPV	CAPEX (capacity installation)	1.16M USD/MW installed power	1.16M USD/MW installed power
Solar – NPV	CAPEX (network connection inland)	590 USD/MW*km	590 USD/MW*km
Solar – NPV	Discount rate	5%	5%
Solar – NPV	Life span	25 years	25 years
Solar – NPV	OPEX	0	0
Solar – NPV	Revenue	Based on the estimated real produced power but using a selling price of 103 USD/MWh (0.103 USD/kWh ~ 0.0767 GBP/kWh)	127TWh*£0.0767/kWh = £9.7bn
Solar – LCOE	Discount rate	8%	8%

(8) R code:

Dropbox:

https://www.dropbox.com/scl/fi/g7sfdkpvv2bo47kceqta/LSHM1_BENV0093.R?rlkey=s2gf4zclcmj9euogqbsbpaj2r&st=3rq0zoay&dl=0

Github:

<https://github.com/amun0905/Solar-suitability-analysis-Indonesia.git>

Some input files are missing from Github due to size constraints

Onedrive: [LSHM1_BENV0093.R](#)

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